Radar range and velocity

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Learning target: Separate a radar’s range measurement from its line-of-sight velocity measurement.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 12.*
Same speed c; the higher-frequency wavelength is 1/3 as large.

02. Worked example: explain the model
Total path 600 m; one-way range 300 m.

03. Read and interpret the model
Delay for range; Doppler frequency shift for radial velocity.

04. Apply the physics
60 m total; 30 m range.

05. Explain your reasoning
No; could move transversely to the line of sight.

06. Review: connect an earlier idea *Review: Unit 6, Day 3.*
35 m/s from $v=f\lambda$.

07. Exit check
500 m.

08. STRETCH: test the model
600 to 450 m; approaching, assuming consistent geometry and timing.

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Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 12.*
$\lambda \approx 0.06$ m; microwave.

02. Worked example: explain the model
Total path 600 m; one-way range 300 m.

03. Read and interpret the model
Delay for range; Doppler frequency shift for radial velocity.

04. Apply the physics
Range 6×10^6 m; radial speed 20 m/s.

05. Explain your reasoning
No; could move transversely to the line of sight.

06. Review: connect an earlier idea *Review: Unit 6, Day 3.*
$v = 30$ m/s; $\lambda = 30/60 \approx 0.5$ m.

07. Exit check
Range describes position along line of sight; Doppler shift describes radial velocity.

08. STRETCH: test the model
-120 m/s; transverse motion is not determined by range change alone.